PR7 Worked Examples: Graphs of Rational Functions

Recording companion for the PR7 CheckIt support video — Math 172, Santa Ana College

How to use this document

- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

The one question behind every example

Which factors cancel, and which ones stay? A factor that appears on both the top and the bottom cancels: it makes a hole, and you find the hole's height by plugging into the cancelled form. A factor left on the bottom makes a vertical asymptote. Factors left on the top make the zeros. Plug in x = 0 for the y-intercept. Compare the degrees of the top and bottom for the horizontal asymptote.

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Example 1: Cancel or Stay?

$$f(x)=(3(x+2)(x-4))/(2(x-4)(x+1))$$

Question

What happens at x = 4 and at x = −1?

- **A.** A hole at x = 4, and a vertical asymptote at x = −1
- **B.** Vertical asymptotes at both x = 4 and x = −1
- **C.** A vertical asymptote at x = 4, and a hole at x = −1
- **D.** A zero at x = 4, and a vertical asymptote at x = −1

Answer and reasoning

$(x-4)$ is on the top and the bottom, so it cancels: $f(x)=(3(x+2))/(2(x+1))$ everywhere except $x=4$, where the original is $0/0$. That is a hole. $(x+1)$ stays on the bottom, so $x=-1$ is a vertical asymptote. Calling both asymptotes misses the cancelling; the swapped version mixes up which factor cancels; and calling 4 a zero misses that $(x-4)$ cancels. **A hole at x = 4, and a vertical asymptote at x = −1.**

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Example 2: Hole Height

$$f(x)=(-(x+5)(x-2))/(3(x-2)(x+4))$$

Question

The graph has a hole at x = 2. What is its y-coordinate?

A. $7/18$
B. 0
C. $-7/18$
D. $-5/12$

Answer and reasoning

Cancel first; away from the hole, the formula is $(-(x+5))/(3(x+4))$. Then plug in 2: $(-(7))/(3(6))=-7/18$. The hole is $(2,-7/18)$. $7/18$ drops the negative in front; 0 comes from reading the original's $0/0$ as 0; and $-5/12$ comes from plugging in 0 instead of 2, which gives the y-intercept. ***−7/18*.**

Example 3: Zeros

$$f(x)=(-4(x-3))/(5(x-3)(x+2)^2)$$

Question

What are the zeros of f?

A. x = 3, from the top factor
B. There are none.
C. x = −2, from the bottom
D. x = −4/5, from the numbers in front

Answer and reasoning

Zeros come from top factors that do not cancel. The only top factor, $(x-3)$, cancels with the bottom, which leaves the formula $(-4)/(5(x+2)^2)$ everywhere except the hole. The top is now just -4, which is never 0. At $x=3$ there is a hole, not a zero; $x=-2$ makes the bottom 0, so it is a vertical asymptote; and $-4/5$ is a ratio of coefficients, not an x-value where f is 0. **There are none.**

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Example 4: Vertical Asymptotes

$$f(x)=(5(x+2)(x-3))/(4(x-3)(x-1)^2)$$

Question

What are the vertical asymptotes of f?

A. x = 1 and x = 3
B. x = −1 only
C. x = 1 and x = −2
D. x = 1 only

Answer and reasoning

Vertical asymptotes come from the bottom factors that are left after cancelling. $(x-3)$ is on the top and the bottom, so it cancels: $x=3$ is a hole, at $(3,25/16)$, not an asymptote. $(x-1)^2$ stays, so $x=1$ is the vertical asymptote. The square does not make a second asymptote; it makes the graph go the same way on both sides of $x=1$, here up, since the cancelled formula $(5(x+2))/(4(x-1)^2)$ is a positive number over a tiny positive number near 1. Together, the asymptote and the hole give the domain: all real numbers except 1 and 3. Adding $x=3$ counts the cancelled factor; $x=-1$ reads the sign as written; and $x=-2$ is a zero, from the top. **x = 1 only.**

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Example 5: Level Off 1

$$f(x)=(-2(x-1)^2(x+3))/(3(x+3)(x+4)^2)$$

Question

What is the horizontal asymptote?

A. $y=1$
B. $y=-2/3$
C. $y=-3/2$
D. $y=0$

Answer and reasoning

The top has degree $2+1=3$ and so does the bottom. Equal degrees: the asymptote is the ratio of the leading coefficients, and those include the numbers in front: $(-2)/3$. $y=1$ compares only the x^3 terms and forgets the -2 and the 3; $-3/2$ puts the ratio upside down; and $y=0$ goes with a bottom of higher degree. ***y=−2/3*.**

Example 6: Level Off 2

$$f(x)=(-3(x-2))/(2(x-2)(x+1))$$

Question

What is the horizontal asymptote?

- **A.** $y=0$
- **B.** $y=-3/2$
- **C.** There is none.
- **D.** $y=-1$

Answer and reasoning

The top has degree 1 and the bottom has degree 2. When the bottom wins, the fraction shrinks toward 0 far out, so the asymptote is $y=0$. $-3/2$ is the ratio of the numbers in front, which matters only when the degrees match; "none" goes with a top of higher degree; and $y=-1$ copies the vertical asymptote's value, $x=-1$, onto the wrong axis. ***y=0*.**

Example 7: Level Off 3

$$f(x)=((x-5)^2(x+1))/(3(x+1)(x-2))$$

Question

What is the horizontal asymptote?

- **A.** $y=1/3$
- **B.** $y=0$
- **C.** x = 2, where the bottom is 0
- **D.** There is none.

Answer and reasoning

The top has degree 3 and the bottom has degree 2, so the top wins and the graph does not level off: there is no horizontal asymptote, which is what CheckIt expects you to say. Far out, the graph follows a slanted line instead; long division, from PR4, finds it. *1/3* would be right only if the degrees matched; *y=0* goes with a bottom of higher degree; and *x=2* is the vertical asymptote. **There is none.**

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Example 8: Cross the y-Axis

$$f(x)=(3(x-1)(x+5))/(4(x+5)(x+3))$$

Question

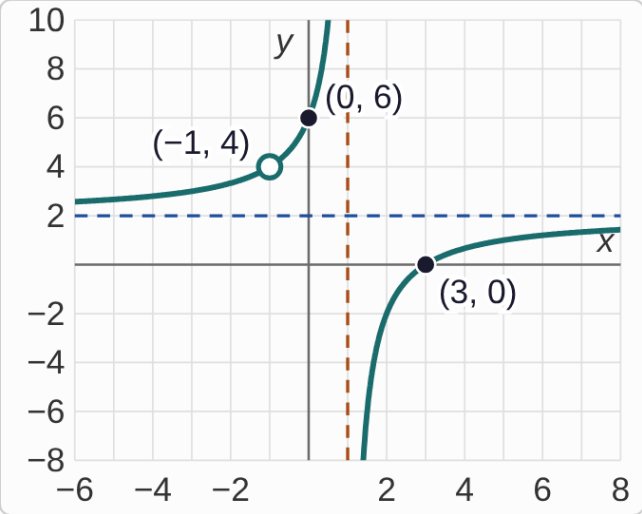
What is the y-intercept?

- **A.** *(0, 1/4)*
- **B.** *(0, 3/4)*
- **C.** *(0, −1/4)*
- **D.** *(0, −15)*

Answer and reasoning

Plug in 0: $f(0)=(3(-1)(5))/(4(5)(3))=(-15)/60=-1/4$. The cancelled form gives the same answer: $(3(-1))/(4(3))=-1/4$. *1/4* drops the negative from *(0−1)*; *3/4* is the horizontal asymptote; and −15 is the top alone, without dividing by the bottom. ***(0,−1/4)*.**

Example 9: Match the Formula



Plotted data: A graph on axes with x from −6 to 8 and y from −8 to 10. A dashed vertical line is at x = 1 and a dashed horizontal line is at y = 2. Left of x = 1, the curve comes in from the left just above y = 2, rises through an open circle at (−1, 4) and through the point (0, 6), and shoots up beside x = 1. Right of x = 1, the curve comes up from the bottom beside x = 1, crosses the x-axis at (3, 0), and levels off just below y = 2. The open circle and both intercepts are labeled.

Question

Which formula matches this graph?

- **A.** $f(x)=(2(x-3)(x-1))/((x-1)(x+1))$
- **B.** $f(x)=((x-3)(x+1))/(2(x+1)(x-1))$
- **C.** $f(x)=(2(x-3)(x+1))/((x+1)(x-1))$
- **D.** $f(x)=(2(x+3)(x+1))/((x+1)(x-1))$

Answer and reasoning

The open circle at $x=-1$ means $(x+1)$ cancels; the asymptote at $x=1$ means $(x-1)$ stays on the bottom; the zero at 3 means $(x-3)$ stays on top; and leveling off at $y=2$ means the numbers in front give $2/1$. Check: the cancelled form $(2(x-3))/(x-1)$ gives 6 at $x=0$ and 4 at $x=-1$, the height of the hole. Cancelling $(x-1)$ instead of $(x+1)$ swaps the hole

and the asymptote; the 2 on the bottom levels off at *1/2*; and *(x+3)* puts the zero at −3. ***f(x)=(2(x−3)(x+1))/((x+1)(x−1))*.**

Example 10: Beside the Asymptote

$$f(x)=(-2(x+4))/(3(x+4)(x-2)^2)$$

Question

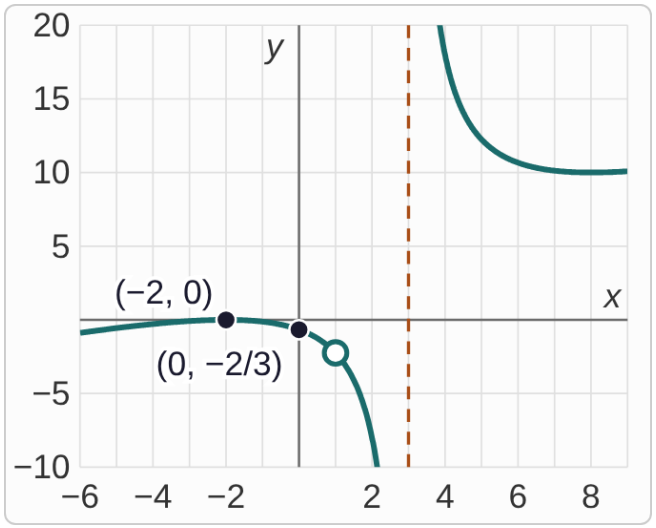
Which describes the graph just to the left and just to the right of x = 2?

- **A.** Down on both sides
- **B.** Up on both sides
- **C.** Down on the left, up on the right
- **D.** Up on the left, down on the right

Answer and reasoning

Cancelled, the formula is $(-2)/(3(x-2)^2)$, away from the hole. Near $x=2$, the bottom $3(x-2)^2$ is a tiny positive number on both sides, because a square is never negative. So f is -2 over a tiny positive number: a huge negative number on both sides. Up on both sides forgets the -2; the two mixed choices would need a factor that changes sign, like $(x-2)$ to the first power. **Down on both sides.**

Example 11: Read the Graph



Plotted data: A graph on axes with x from −6 to 9 and y from −10 to 20. A dashed vertical line is at x = 3; no other dashed lines are drawn. Left of x = 3, the curve rises from the lower left to just touch the x-axis at (−2, 0), turns down through (0, −2/3) and an open circle at x = 1, a little below the x-axis, and drops beside x = 3. Right of x = 3, the curve comes down from the top, bottoms out near (8, 10), and then rises again to the right. The two intercepts are labeled.

$$f(x)=((x+2)^2(x-1))/(2(x-1)(x-3))$$

Question

Which list matches the graph and the formula?

- **A.** Hole (1, 0); asymptote x = 3; no horizontal asymptote; zero −2
- **B.** Hole (1, −9/4); asymptote x = 3; horizontal asymptote y = 1/2
- **C.** Hole (1, −9/2); asymptote x = 3; no horizontal asymptote; zero −2
- **D.** Hole (1, −9/4); asymptote x = 3; no horizontal asymptote; zero −2

Answer and reasoning

$(x-1)$ cancels, so there is a hole at $x=1$, with height $((1+2)^2)/(2(1-3))=-9/4$. $(x-3)$ stays on the bottom: the asymptote $x=3$. The top has degree 3 and the bottom degree 2, so the graph never levels off; far out it heads up on the right and down on the left. The

zero is −2, where the squared factor makes the graph touch the axis and turn. A height of 0 reads the original's *0/0* as 0; *−9/2* forgets the 2 on the bottom; and *y=1/2* would need equal degrees. **Hole *(1,−9/4)*; asymptote *x=3*; no horizontal asymptote; zero −2.**

Example 12: Sketch It

$$f(x)=(4(x+3)(x-2))/(2(x-2)(x-1))$$

Question

CheckIt will ask you to sketch this. The graph crosses the x-axis at −3 and passes through (0, −6). Reading from left to right, which description is right?

A. Near y = 2, down beside x = 1; then from the top, a hole at (2, 10), back toward y = 2

B. Near y = 2, up beside x = 1; then from the bottom, a hole at (2, 10), back toward y = 2

C. Near y = 0, down beside x = 1; then from the top, a hole at (2, 10), back toward y = 0

D. Near y = 2, down beside x = 1; then from the top, down beside x = 2, back toward y = 2

Answer and reasoning

Cancelled, the formula is $(2(x+3))/(x-1)$, away from the hole. Equal degrees with $4/2=2$ in front: the graph levels off at $y=2$. The zero is −3, and $f(0)=(2(3))/(-1)=-6$. Just left of 1, the bottom is a tiny negative number, so f dives down; just right of 1, it is a tiny positive number, so f comes down from the top. The hole is at $(2,(2(5))/1)=(2,10)$. Going up beside 1 on the left gets the sign of $x-1$ wrong; $y=0$ forgets the equal degrees; and a second asymptote at 2 misses the cancelling. **Near y = 2, down beside x = 1, then from the top, a hole at (2, 10), back toward y = 2.**

Example 13: Slip 1

$$f(x)=(-(x-5)(x+2))/(4(x+2)(x-3))$$

Question

A student says f has vertical asymptotes at x = −2 and x = 3. What is the mistake, if any?

A. There is no mistake.
B. The signs are flipped: the asymptotes are really at x = 2 and x = −3.
C. (x + 2) cancels, so x = −2 is a hole; the only asymptote is x = 3.
D. x = 5 is also a vertical asymptote, from the factor (x − 5) on top.

Answer and reasoning

$(x+2)$ is on the top and the bottom, so it cancels, and $x=-2$ is a hole at $(-2,-7/20)$. Only $(x-3)$ stays on the bottom, so $x=3$ is the only vertical asymptote. The signs were right, and $(x-5)$ is on top, so 5 is a zero, not an asymptote. **(x + 2) cancels: a hole at x = −2, and the only asymptote is x = 3.**

Example 14: Slip 2

$$f(x)=(4(x+1)(x-3))/(5(x-3)(x+2))$$

Question

A student says the horizontal asymptote is y = 1, since the top and bottom both have degree 2. What is the mistake, if any?

A. There is no mistake.
B. The numbers in front count: the asymptote is y = 4/5.
C. With equal degrees, the asymptote is always y = 0, the x-axis.
D. There is none, since the factor (x − 3) cancels.

Answer and reasoning

Equal degrees is the right thing to check, but the leading coefficients include the 4 and the 5: multiplied out, the top starts $4x^2$ and the bottom starts $5x^2$. So the asymptote is $y=4/5$. $y=0$ goes with a bottom of higher degree, and cancelling a factor makes a hole without changing the asymptote. **The asymptote is $y=4/5$.**

Example 15: Slip 3

$$(2(x-4))/(3(x-1))$$
$$(2(-7))/(3(-4))=7/6$$

Question

For $f(x)=(2(x-4)(x+3))/(3(x+3)(x-1))$, a student wrote the cancelled form (first line), plugged in −3 (second line), and said the hole is (−3, 7/6). What is the mistake, if any?

A. There is no mistake.
B. The hole's height is 0, since (x + 3) is 0 at x = −3.
C. The hole is at x = 1, where the bottom is 0.
D. Two negatives multiply to a negative, so it is −7/6.

Answer and reasoning

$(x+3)$ cancels, so the hole is at $x=-3$, and its height comes from the cancelled form: $(2(-7))/(3(-4))=(-14)/(-12)=7/6$. In the original form you would get $0/0$, not 0. $x=1$ is the vertical asymptote, and a negative times a negative is positive. **There is no mistake.**

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Example 16: Slip 4

$$f(x)=(5(x+1))/(2(x+1)(x-4)^2)$$

Question

A student says the zero of f is x = −1, from the factor (x + 1) on top. What is the mistake, if any?

A. There is no mistake.
B. The zero is x = 4, from the squared factor $(x - 4)^2$ on the bottom.
C. The zero is x = 1, since the factor (x + 1) means x is 1.
D. (x + 1) cancels, so x = −1 is a hole and f has no zeros.

Answer and reasoning

$(x+1)$ is on the top and the bottom, so it cancels, leaving the formula $5/(2(x-4)^2)$ away from the hole. The top is now 5, never 0, so there are no zeros, and $x=-1$ is a hole at $(-1,1/10)$. $x=4$ makes the bottom 0, so it is a vertical asymptote, and $(x+1)$ goes with −1, not 1. **There are no zeros; x = −1 is a hole.**

Example 17: Slip 5

$$f(x)=(-3(x+4)^2(x-2))/(5(x-2)(x+1)^2)$$

Question

A student says the horizontal asymptote is y = 0, because the bottom has a squared factor. What is the mistake, if any?

A. There is no mistake.
B. There is none, since the top also has a squared factor.
C. Both have degree 3, so the asymptote is y = −3/5.
D. The asymptote is y = −1, from the two squared factors.

Answer and reasoning

What matters is the total degree, not one squared factor. The top has degree $2+1=3$ and the bottom has degree $1+2=3$: equal, so the asymptote is the ratio of the leading coefficients, $y=-3/5$. Having a square on top does not remove the asymptote, and the leading coefficients are -3 and 5, not the factors' 1s. **Equal degrees: $y=-3/5$.**